CTMC

state i, j

v_i rate at which the process makes a transition when in state i

P_{ij} prob. that this transition is into state j

failure rate $\lambda(t) = -\frac{d}{dt} \ln \bar{F}(t)$

$$= \frac{f(t)}{\bar{F}(t)} = \frac{\lambda \cancel{e^{-\lambda t}}}{\cancel{e^{-\lambda t}}}$$

q_{ij} rate, when in state i, at which the process makes a transition into state j

$$q_{ij} = v_i P_{ij}$$

$$\sum_j q_{ij} = v_i \underbrace{\sum_j P_{ij}}_{=1} = v_i$$

$P_{ij}(t) = P(X(t) = j \mid X(0) = i)$

Lemma

(a) $$\lim_{h \to 0} \frac{1 - P_{ii}(h)}{h} = v_i$$

(b) $$\lim_{h \to 0} \frac{P_{ij}(h)}{h} = q_{ij} \quad \text{when } i \neq j$$

Sol.

(a) $$1 - P_{ii}(h) = v_i h + o(h)$$

$$\lim_{h \to 0} \frac{1 - P_{ii}(h)}{h} = v_i$$

(b) $$P_{ij}(h) = q_{ij} h + o(h)$$

$$\lim_{h \to 0} \frac{P_{ij}(h)}{h} = q_{ij}$$

—x—

$$P_{ij}(t+s) = P(X(t+s)=j \mid X(0)=i)$$

$$= \sum_k P(X(t+s)=j, X(t)=k \mid X(0)=i)$$

$$= \sum_k P(X(t+s)=j \mid X(t)=k, \cancel{X(0)=i})$$

$$P(X(t)=k \mid X(0)=i)$$

$$= \sum_k P_{kj}(s) \, P_{ik}(t)$$

$$\boxed{P_{ij}(t+s) = \sum_k P_{ik}(t) \, P_{kj}(s)}$$

Chapman Kolmogorov equation.